Software Project and Management

Assignment 3 (Third Deliverable)

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SE-E

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Task 1: Timeline Estimation and Scheduling of WBS Items:

Introduction

This section provides the timeline estimates, start and finish dates, dependencies, and responsible team members for each WBS item of the *Nutrition Coach Agent* project. The schedule is based on the Assignment 02 WBS and Gantt Chart.

Assumptions

- Project timeline window: 14 Sep 2025 – 12 Dec 2025 (semester timeline).
- Dates are calendar days; if needed, convert to working days with the WORKDAY formula.
- Each team member contributes about 20 hours per week.
- Durations already include small buffers; 10 % contingency recommended for budgeting.

Timeline Table:

Task ID	Task Name	Duration (days)	Start Date	Finish Date	Dependencies	Assigned To	Justification
T1	Project Charter	7	14-Sep-2025	20-Sep-2025	—	Azghan Ahmad	Prepare official project charter and obtain initial approvals.
T2	Stakeholder Identification	3	20-Sep-2025	22-Sep-2025	T1	Azghan Ahmad	Identify stakeholders and their interests for alignment.
T3	Team Formation	2	22-Sep-2025	23-Sep-2025	T1	Azghan Ahmad	Define roles and responsibilities among team members.
T4	Define Scope Statement	3	01-Oct-2025	03-Oct-2025	T2	Azghan Ahmad	Create scope statement, assumptions and constraints.
T5	Requirements Collection	3	02-Oct-2025	04-Oct-2025	T4	Rabail	Collect functional and non-functional requirements.
T6	UI/UX Design	5	11-Oct-2025	15-Oct-2025	T5	Amna Asif	Design wireframes and page layouts for the Nutrition Coach.
T7	Database Schema Design	4	15-Oct-2025	18-Oct-2025	T5	Rabail	Design SQLite schema to support meals, users, logs.

T8	System Architecture	5	18-Oct-2025	22-Oct-2025	T5	Rabail	Define modules, APIs, data flow, and integration points.
T9	Synthetic Dataset Creation	8	20-Oct-2025	27-Oct-2025	T5	Rabail	Generate synthetic nutrition dataset for testing.
T10	Backend Development	8	01-Nov-2025	08-Nov-2025	T7; T8; T9	Rabail	Implement Flask backend endpoints and core logic.
T11	Recommendation Engine (Rules)	5	08-Nov-2025	12-Nov-2025	T9; T10	Rabail	Implement rule-based recommendation logic and tests.
T12	Frontend Development	11	05-Nov-2025	15-Nov-2025	T6; T8	Amna Asif	Build responsive UI for logging, charts, and recommendations.
T13	Database Integration	3	12-Nov-2025	14-Nov-2025	T10; T7	Rabail	Connect backend with SQLite for persistence.
T14	Unit Testing	3	16-Nov-2025	18-Nov-2025	T10; T12	All Members	Write and run unit tests for backend and frontend modules.
T15	API Integration (Supervisor Agent)	4	21-Nov-2025	24-Nov-2025	T10; T11	Amna Asif	Integrate Nutrition Coach API with Supervisor Agent.
T16	Integration Testing	5	24-Nov-2025	28-Nov-2025	T13; T14; T15	All Members	End-to-end integration tests and validation of data flows.
T17	Bug Fixes & Refinement	5	28-Nov-2025	02-Dec-2025	T16	Rabail	Resolve defects found during integration testing.
T18	User Acceptance Testing (UAT)	3	02-Dec-2025	04-Dec-2025	T17	Azghan Ahmad	Conduct UAT sessions and record feedback for acceptance.
T19	Deployment Build	1	06-Dec-2025	06-Dec-2025	T18	Rabail	Package application for demo deployment.
T20	Upload to Demo Environment	2	06-Dec-2025	07-Dec-2025	T19	Amna Asif	Upload demo build to live environment or cloud.
T21	User Guide & Documentation	2	07-Dec-2025	08-Dec-2025	T17; T18	Azghan Ahmad	Prepare user manual, installation, and API documentation.

T22	Handover Demo	1	08-Dec-2025	08-Dec-2025	T20; T21	All Members	Final demo for instructor and integration coordinator.
T23	Feedback Collection	1	09-Dec-2025	09-Dec-2025	T22	Azghan Ahmad	Collect post-demo feedback and improvement notes.
T24	Recommendation Accuracy Review	1	09-Dec-2025	10-Dec-2025	T22	Rabail	Review model accuracy and adjust rules if needed.
T25	Final Report Compilation	1	11-Dec-2025	11-Dec-2025	T21; T22; T23	Azghan Ahmad	Compile final report and required artifacts.
T26	Sign-offs & Archive	1	12-Dec-2025	12-Dec-2025	T25	Azghan Ahmad	Obtain formal sign-offs and archive project artifacts.

Justification for Major Work Packages

- **Project Charter & Initiation (T1–T3):** Short administrative tasks for stakeholder alignment and approvals.
- **Requirements, Design & Dataset (T4–T9):** Time for requirement gathering, UI/UX design, database schema, and synthetic dataset creation.
- **Implementation & Engine (T10–T13):** Backend, recommendation engine, and frontend tasks are core development activities; durations reflect complexity and integration needs.
- **Testing & Integration (T14–T18):** Dedicated period for unit tests, API integration, integration testing, and bug fixes.
- **Deployment & Handover (T19–T22):** Packaging, upload, demo, and documentation for submission.

Dependencies and Sequencing

System architecture and data availability must precede implementation tasks.
Backend and frontend work must finish before integration testing.
Testing and bug fixing happen before UAT and deployment.
Documentation and final report come after integration and acceptance.

Risks and Mitigations

- Data delays → start dataset creation early.
- Integration issues → define API contracts and test incrementally.
- Team availability → weekly meetings and shared task board.
- Add 10 % buffer in schedule and budget for uncertainties.

Task 2: Network Diagram (AON) and Slack Analysis:

Step 1: Forward Pass (ES & EF):

Rules:

- ES = max (EF of all predecessors)
- EF = ES + Duration

Task	Duration	Predecessor(s)	ES	EF
T1	7	–	0	7
T2	3	T1	7	10
T3	2	T1	7	9
T4	3	T2	10	13
T5	3	T4	13	16
T6	5	T5	16	21
T7	4	T5	16	20
T8	5	T5	16	21
T9	8	T5	16	24
T10	8	T7, T8, T9	24	32
T11	5	T10, T9	32	37
T12	11	T6, T8	21	32
T13	3	T10, T7	32	35
T14	3	T10, T12	32	35
T15	4	T10, T11	37	41
T16	5	T13, T14, T15	41	46
T17	5	T16	46	51
T18	3	T17	51	54
T19	1	T18	54	55
T20	2	T19	55	57
T21	2	T17, T18	51	53
T22	1	T20, T21	57	58
T23	1	T22	58	59
T24	1	T22	58	59
T25	1	T21, T22, T23	59	60
T26	1	T25	60	61

.

Step 2: Backward Pass (LS & LF):

Rules:

- LF of last task = EF of last task (project end) = 61
- LS = LF − Duration
- LF of predecessors = min (LS of all successors)

Task	Duration	Successor(s)	LF	LS
T26	1	–	61	60
T25	1	T26	60	59
T24	1	T25	59	58
T23	1	T25	59	58
T22	1	T23, T24, T25	58	57
T21	2	T22, T25	57	55
T20	2	T22	57	55
T19	1	T20	55	54
T18	3	T19, T21	54	51
T17	5	T18, T21	51	46
T16	5	T17	46	41
T15	4	T16	41	37
T14	3	T16	41	38
T13	3	T16	41	38
T12	11	T14	38	27
T11	5	T15	37	32
T10	8	T11, T13, T14, T15	32	24
T9	8	T10, T11	24	16
T8	5	T10, T12	24	19
T7	4	T10, T13	24	20
T6	5	T12	27	22
T5	3	T6, T7, T8, T9	16	13
T4	3	T5	13	10
T3	2	–	13	11
T2	3	T4	10	7
T1	7	T2, T3	7	0

Step 3: Total Slack & Free Slack:

- **Total Slack (TS) = LS − ES = LF − EF**
- **Free Slack (FS) = min (ES of successors) − EF**

Task	TS	FS	Notes
T1	0	0	Critical
T2	0	0	Critical
T3	4	2	Non-critical
T4	0	0	Critical
T5	0	0	Critical
T6	6	1	Non-critical
T7	0	0	Critical
T8	3	0	Non-critical
T9	0	0	Critical
T10	0	0	Critical
T11	0	0	Critical
T12	5	0	Non-critical
T13	3	0	Non-critical
T14	6	0	Non-critical
T15	0	0	Critical
T16	0	0	Critical
T17	0	0	Critical
T18	0	0	Critical
T19	0	0	Critical
T20	0	0	Critical
T21	4	0	Non-critical
T22	0	0	Critical
T23	0	0	Critical
T24	0	0	Critical
T25	0	0	Critical
T26	0	0	Critical

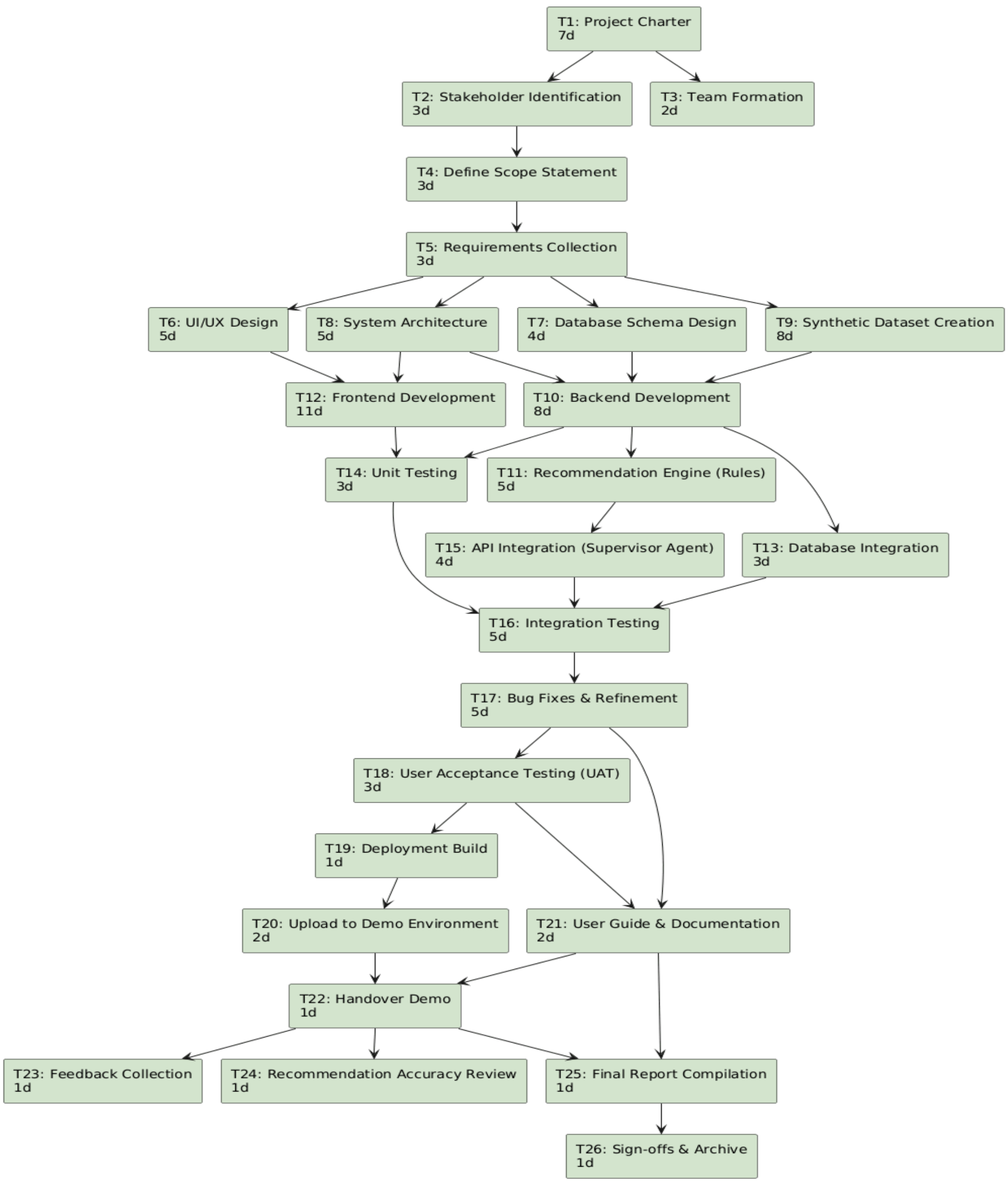
Step 4: Correct Critical Path:

Only tasks with TS = 0:

T1 → T2 → T4 → T5 → T7 → T9 → T10 → T11 → T15 → T16 → T17 → T18 → T19 → T20 → T22 → T23/T24 → T25 → T26

- Tasks like T3, T6, T8, T12, T13, T14, T21 are **parallel/non-critical**, some with slack.

Step 5: AON Diagram:



Step 6: Interpretation:

“The network diagram illustrates the logical sequence of project tasks. The critical path consists of tasks T1 → T2 → T4 → T5 → T7 → T9 → T10 → T11 → T15 → T16 → T17 → T18 → T19 → T20 → T22 → T23/T24 → T25 → T26, determining the minimum project duration of 61 days. Tasks T3, T6, T8, T12, T13, T14, and T21 have positive slack, meaning they can be delayed without affecting overall project completion. Parallel paths exist, allowing concurrent execution of non-critical tasks, improving resource utilization. Total Slack and Free Slack calculations provide insights into schedule flexibility for risk management

Task 3: Cost Estimation, Budgeting (Excel), and Earned Value Analysis:

Step 1: Cost Estimation & Budget Spreadsheet:

	A	B	C	D	E	F
1	**Item No**	**Cost Item**	**Quantity**	**Unit Cost (PKR)**	**Subtotal (PKR)**	**Notes**
2	1	Project Charter & Kick-off meeting	1	30,000	30,000	Documentation, approvals
3	2	Stakeholder / Team Workshops	2	20,000	40,000	Meetings + refreshments
4	3	Requirements Gathering & Analysis	1	50,000	50,000	Functional + non-functional requirements
5	4	UI/UX Design	1	60,000	60,000	Wireframes, mockups
6	5	Database Schema & Architecture	1	40,000	40,000	DB + API design
7	6	Backend Development	1	100,000	100,000	Flask backend modules
8	7	Frontend Development	1	90,000	90,000	React/HTML/CSS UI
9	8	Synthetic Data Generation	1	30,000	30,000	Data creation for testing
10	9	Testing (Unit + Integration)	1	50,000	50,000	Test planning, execution
11	10	Deployment & Demo Setup	1	25,000	25,000	Packaging + demo environment
12	11	Documentation & User Guide	1	20,000	20,000	Manuals, installation doc
13	**Subtot**	SubTotal			**535,000**	
14	Conting ency (10%)	Contigency (10%)		53,500	Risk buffer	
15	**Total Projec**	Total Project Budget (BAC)		**588,500**	Budget at Completion	

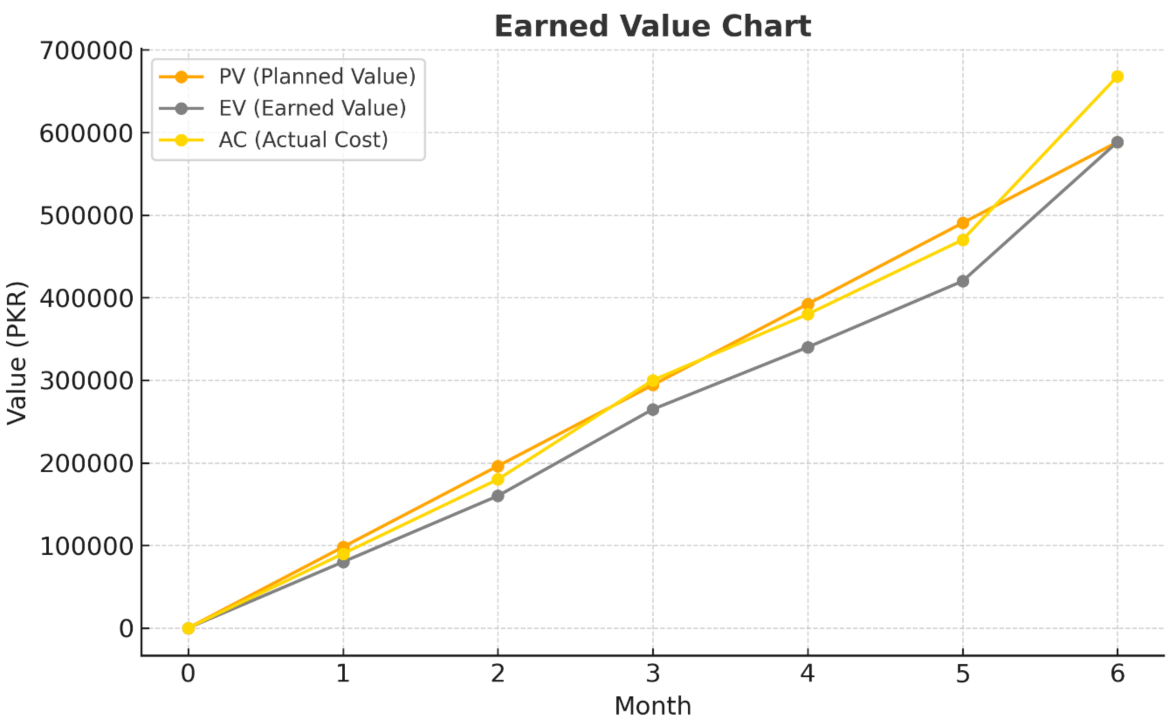
Step 2: Assumptions after 3 months:

Assuming 6-month project duration, after 3 months:

- **Planned Value (PV)** = 50% × BAC = 0.5 × 588,500 = **294,250 PKR**
- **Earned Value (EV)** = 45% × BAC = 0.45 × 588,500 = **264,825 PKR**
- **Actual Cost (AC)** = **300,000 PKR** (spent so far)

Term	Formula	Value	Interpretation
CV	EV − AC	264,825 − 300,000 = −35,175	Over budget
SV	EV − PV	264,825 − 294,250 = −29,425	Behind schedule
CPI	EV / AC	264,825 / 300,000 ≈ 0.88	Cost efficiency <1 → over budget
SPI	EV / PV	264,825 / 294,250 ≈ 0.90	Schedule efficiency <1 → behind schedule
EAC	BAC / CPI	588,500 / 0.88 ≈ 668,068	Total projected cost if current trend continues
Estimated Duration	OriginalDuration / SPI	6 / 0.90 ≈ 6.67 months	Project finish delayed ~20 days

Step 3: Earned Value Chart:



Month	PV (Planned Value)	EV (Earned Value)	AC (Actual Cost)
0	0	0	0
1	98,083	80,000	90,000
2	196,167	160,000	180,000
3	294,250	264,825	300,000
4	392,333	340,000	380,000
5	490,417	420,000	470,000
6	588,500	588,500	668,068

Step 4: Interpretation:

"The Earned Value (EV) chart shows the project's PV, EV, and AC over time. After three months, EV is below PV, indicating the project is **behind schedule** (SPI < 1). AC is above EV, showing the project is **over budget** (CPI < 1). If trends continue, the total cost at completion (EAC) is projected higher than BAC, and the project duration will slightly exceed the original estimate."